

Experiment-12

Python Program for Tic Tac Toe Game

Aim

To develop a Python program to implement a **Tic-Tac-Toe game** for two players.

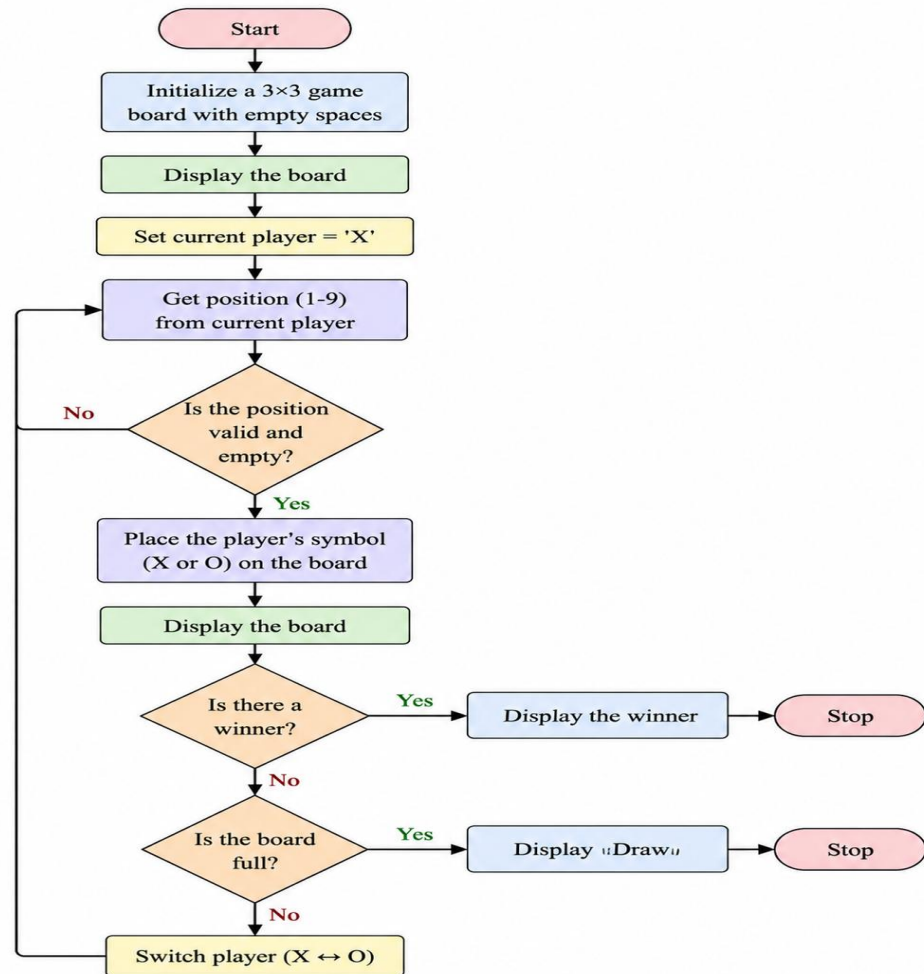
Objective

- To implement a two-player Tic-Tac-Toe game using Python.
- To check the winning conditions after each move.
- To display the winner or draw result.

Algorithm

1. Start the program.
2. Initialize a 3×3 game board.
3. Display the board.
4. Select the current player (X or O).
5. Get the position from the player.
6. Check whether the position is empty.
7. Place the player's symbol on the board.
8. Check for a winning combination.
9. If a player wins, display the winner and stop.
10. If all positions are filled, display **Draw**.
11. Otherwise, switch the player and repeat.
12. Stop the program.

Flowchart:



Code:

```
board = [' '] * 9
```

```
def show():
```

```
    print(board[0], '|', board[1], '|', board[2])
```

```
    print('--+---+--')
```

```
    print(board[3], '|', board[4], '|', board[5])
```

```
    print('--+---+--')
```

```
    print(board[6], '|', board[7], '|', board[8])
```

```
def win(p):
```

```
    w = [(0,1,2),(3,4,5),(6,7,8),
```

```
          (0,3,6),(1,4,7),(2,5,8),
```

(0,4,8),(2,4,6)]

return any(all(board[i] == p for i in x) for x in w)

for turn in range(9):

show()

p = 'X' if turn % 2 == 0 else 'O'

pos = int(input(f'Player {p}, enter position (1-9): ')) - 1

if board[pos] != '':

print("Invalid move!")

continue

board[pos] = p

if win(p):

show()

print("Player", p, "wins!")

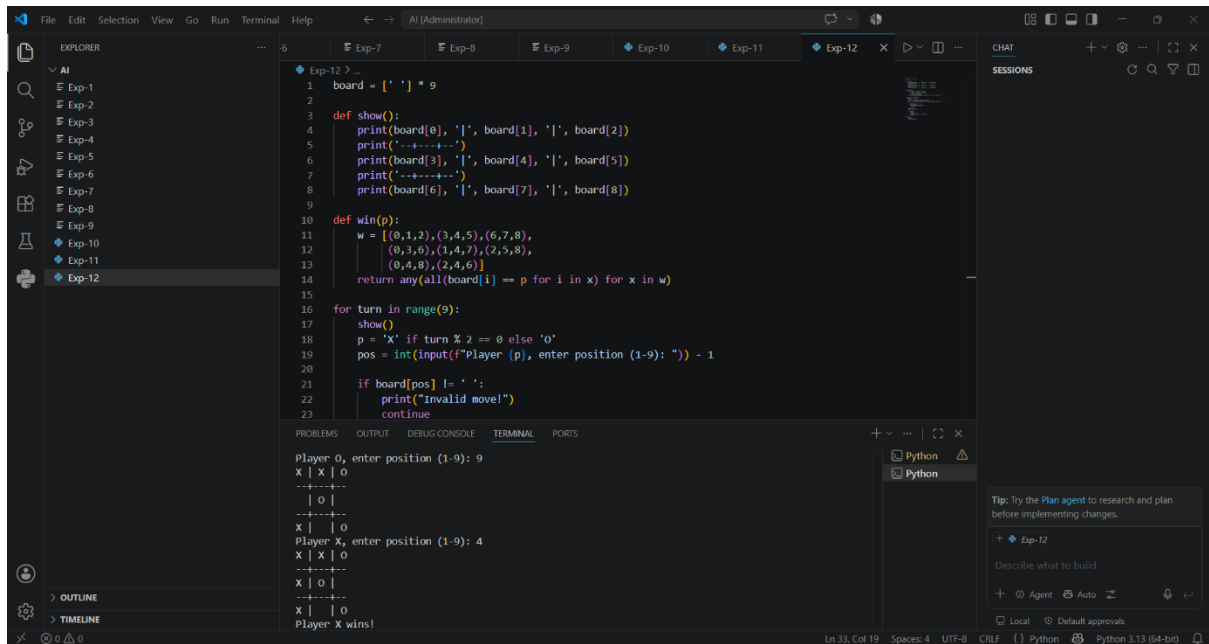
break

else:

show()

print("Draw!")

Output:



```
1 board = [' ']*9
2
3 def show():
4     print(board[0], '|', board[1], '|', board[2])
5     print('---+---+---')
6     print(board[3], '|', board[4], '|', board[5])
7     print('---+---+---')
8     print(board[6], '|', board[7], '|', board[8])
9
10 def win(p):
11     w = [(0,1,2),(3,4,5),(6,7,8),
12          (0,3,6),(1,4,7),(2,5,8),
13          (0,4,8),(2,4,6)]
14     return any(all(board[i] == p for i in x) for x in w)
15
16 for turn in range(9):
17     show()
18     p = 'X' if turn % 2 == 0 else 'O'
19     pos = int(input(f"Player {p}, enter position (1-9): ")) - 1
20
21     if board[pos] != ' ':
22         print("Invalid move!")
23         continue
```

Player O, enter position (1-9): 9
X | X | O
---+---+---
| O |
---+---+---
X | | O
Player X, enter position (1-9): 4
X | X | O
---+---+---
X | O |
---+---+---
X | | O
Player X wins!

Result

The Tic-Tac-Toe game was successfully implemented in Python, and the program correctly identifies the winner or draw.

Conclusion

The Tic-Tac-Toe game was successfully developed using Python. The program demonstrates the use of lists, functions, loops, conditions, and user input to implement a simple game.